

A Preview of Computational Animal Linguistics

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Computational animal linguistics is an emerging field that studies non-human animal vocal communication as a language system using data-driven computational approaches. This survey starts with humanity's historical endeavor to understand animal communication and a background on the essential linguistic properties in human languages. It presents current evidence of animals possessing some of these properties previously thought to be unique to humans. Driven by advancements in deep learning and artificial intelligence, computational methods from natural language processing and large language modeling are playing a major role in processing and understanding animal communication on a larger scale moving forward beyond traditional approaches.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Machine learning approaches**; • **Applied computing** → *Computational biology*.

Additional Key Words and Phrases: Computational Animal Linguistics, Animal Communication, Bioacoustics

ACM Reference Format:

Tuan M. Dang, Atrey Desai, Hridayesh Lekhak, Tirza Panunto, Lindsay Pike, Theron S. Wang, and Kenny Q. Zhu. 2018. A Preview of Computational Animal Linguistics. 1, 1 (September 2018), 38 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Animal behavior and communication have been a subject of interest since the beginning of human history [147]. Animal communication in particular has been studied more extensively in recent years, often under the scope of bioacoustics. Bioacoustics pertains to the study of emitted sounds by living organisms, often studied through more traditional and biological methodology. Usually, this is done through observational studies, either in the wild or in laboratories, observing for behavior and categorizing perceived vocalizations to match said behaviors. In particular,

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models, such as the “code model”, which interpret animal information as general signals to be encoded and decoded, have been utilized. More recently, however, there has been pushback against using these methodologies in studying animal communication because of how vastly different it is from how researchers study and come to understand human language, automatically separating the animal’s form of communication from the amount of complex information or structure that can be extracted [43].

To alleviate these concerns and diversify the methodology in which animal communication is studied, linguistics has been brought into the discussion [22]. This is a highly contested issue, as linguistics has historically been considered to be strictly pertaining to human language [51]. Human languages differ from animal communication in structure complexity, neurological capacity, and anatomical structures [8, 84]. That being said, it is difficult to understand the broad complexity of animal communication, particularly considering that animals utilize a broad range of modalities, including visual, tactile, audible, olfactory, and even electrical signals, many of which are beyond natural human capacities to perceive, creating many barriers in understanding their communication systems.

Despite these barriers, interspecific communication exists, often recorded between humans and other animal species. Humans have interacted with dogs and wolves for thousands of years, developing vocal, visual, and tactile communication [209]. These interactions are interesting to note since they often vary from their communicative interactions with members of the same species, indicating communicative flexibility as well as the potential innateness of communication in the animal kingdom, along with a broader complexity to animal communication than currently understood. Other animals, including cats [254], birds [6], domesticated ranching animals [265], and even marine mammals [223], have also developed ways for interspecific communication with humans, also often utilizing a variety of modalities, though there is still much debate on animals’ capacity for language.

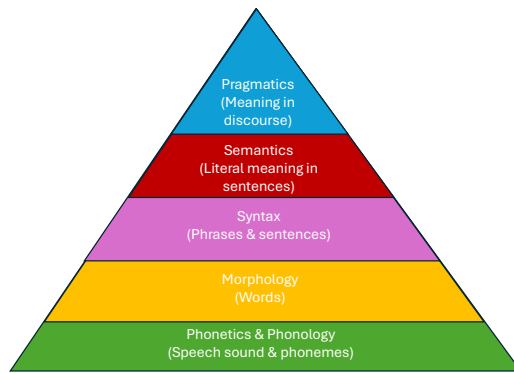


Fig. 1. The hierarchical levels of linguistic analysis, from foundational sounds to complex, context-dependent meaning.

Language, as it is currently considered, has been recorded and defined through the capabilities and structures of human language, rudimentarily through the five basic categories of linguistics: *phonetics and phonology*, *morphology*, *syntax*, *semantics*, and *pragmatics* [166] (see Figure 1). Many argue that because no animal species has yet to match these capabilities and structures, animals do not have language. Others counter-argue that researchers lack the necessary tools to understand animal communication in its entirety, and critique the way language is understood as perhaps too narrow and should be broadened to fully understand language and general communication from an evolutionary standpoint, rather than a unique human characteristic [114]. They also argue that some of the topics listed as characteristics of

language could have evolved from humans for non-communicative purposes, and should be reconsidered more extensively. Though these researchers have yet to come to an agreement or fully concede to oppositional claims, animal communication, studied under the parameters of linguistics, has grown increasingly prevalent [22], predominantly through computational methodology. The historical development from early biological interests to the recent integration of computational techniques is outlined in the timeline of animal communication studies (see Figure 2). In this survey, we are not claiming that there is already conclusive evidence to confirm that animals have language yet. One of our

main goals is to review the studies that have attempted to answer this very question and set a foundation to invite other researchers to join in the discussion.

Computational methodology was first introduced to linguistics in order to understand how language is processed and functions in the human brain, utilized to represent various theories and compare the observational or neurological data from human participants, creating many breakthroughs for the field [167]. With the concept of animal linguistics emerging, animals began undergoing studies utilizing these same language models apart from their vocalizations and context, like in bioacoustics, but from their more complex communication structures, like in linguistics [218]. Computational methodology has also been able to extract information and communication signals that are beyond human perception. These methods are particularly effective with acoustic and vocal modalities. Being able to process, annotate, and train algorithms to encode and analyze animal communication has not only helped researchers understand the dynamics of the phones being produced, but has also taken an abundance of data, quickly and effectively matching and annotating it to produce basic structure within contexts. This has led to many breakthroughs within the field, and computational methodology has been established as an important tool for both traditional and animal linguistics.

To map the diverse methodologies employed in this evolving field, we categorize approaches based on their analytical focus (Bioacoustics vs. Animal Linguistics) and their reliance on automation (Manual vs. AI-Driven Approach), as

Timeline of Animal Communication Studies

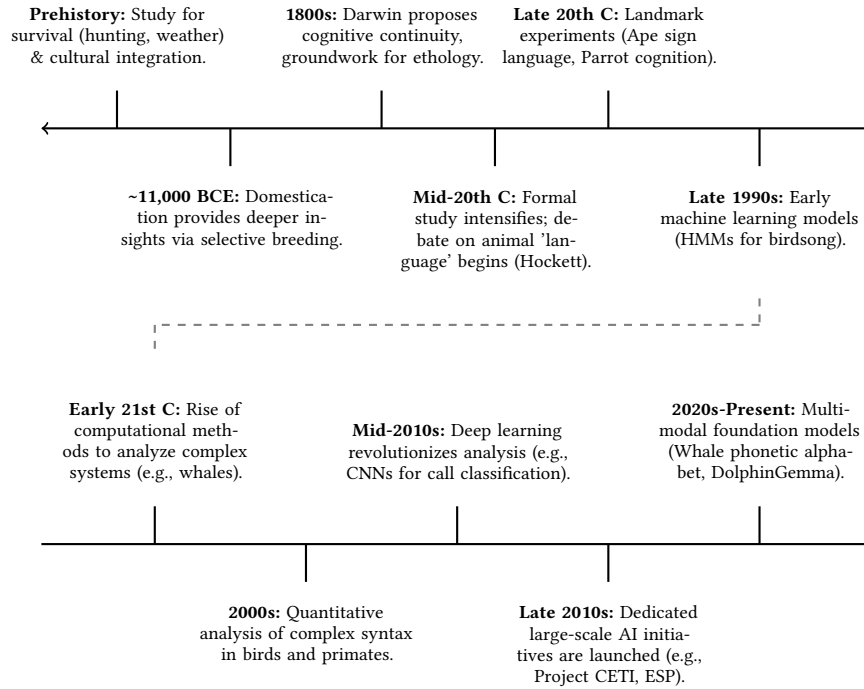


Fig. 2. Key milestones in the history of human interest and study of animal communication. The top timeline provides broad historical context, while the bottom details the recent shift toward computational methods and AI, expanding on the developments of the late 20th and early 21st centuries.

illustrated in Figure 3. This survey primarily concentrates on the **AI-Driven Animal Linguistics** quadrant, exploring advanced techniques like unsupervised unit discovery, phonetic discovery, and grammar induction to uncover latent structures within communication systems. However, these higher-level linguistic investigations are not standalone; they are built upon a pipeline that requires robust preprocessing. Therefore, we also include a discussion of technologies falling under **AI-Driven Bioacoustics**, such as automated detection, automated classification, and source separation. These data-driven processing steps are fundamental prerequisites for effectively isolating communication signals, which is crucial before any investigation into linguistic features can begin.

Literature selection. Papers were identified by searching Web of Science, PubMed, IEEE Xplore, the ACM Digital Library, arXiv, and the ACL Anthology for work published between 1970 and 2026. Queries combined three concept blocks—animal communication terminology; taxonomic terms scoped to a communication context; and computational, linguistic, or behavioral-experimental methods—and were complemented by backward and forward snowballing from key papers and by domain expertise spanning bioacoustics, computational linguistics, and animal behavior. Retrieved records were de-duplicated by DOI and normalized title, and were then manually screened on title and abstract against explicit criteria. We included studies of non-human animal communication

that either apply a computational method to communication data or establish a linguistic design feature on which our hierarchy depends, and that report an identifiable method with empirical grounding. We excluded work on human speech and language alone, sensor and platform engineering without analysis of communicative content, and population-monitoring studies in which vocalizations serve only as a detection proxy rather than an object of analysis. Full texts of the retained records were then assessed, and the studies discussed in this survey are those retained by this process. The complete query strings, the full criteria, and the screening log are provided in the supplementary material¹.

In this paper, Sec. 2 gives an overview of traditional approaches to exploring animal linguistics, categorizing prior research according to the hierarchy of linguistic design features. Sec. 3 outlines a comprehensive five-stage computational pipeline, detailing the progression from data collection and pre-processing to sequence representation, meaning identification, and vocal generation. Sec. 4 presents significant case studies on songbirds, marine mammals, and canines, illustrating how these methodologies have successfully decoded complex communicative features ranging from individual identity to syntactic structure. Sec. 5 synthesizes potential applications, discussing how these computational advancements can transform human–animal interaction, veterinary welfare, and large-scale conservation monitoring. Sec. 6 addresses the significant technical, ethical, and theoretical challenges facing the field, from data scarcity and annotation bias to the fundamental debate over applying human linguistic frameworks to non-human species. Finally, Sec. 7 analyzes current publication trends and identifies future directions in the literature.

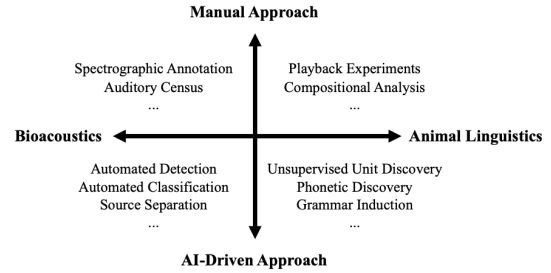


Fig. 3. A methodological framework for animal communication research categorizing approaches by analytical focus (Bioacoustics vs. Animal Linguistics) and implementation strategy (Manual vs. AI-Driven).

¹<https://github.com/UTA-ACL2/cal-animal-comm-corpus>

2 Traditional Approaches to Exploring Animal Linguistics

In this paper, we categorize thirteen linguistic properties [43, 112, 119] that were previously thought to be unique to humans into four levels, with the first level containing the more rudimentary features, up to the fourth level, which has the most complex features (for complete definitions, see Table 1). Features in level one, signal features, relate to how sound is emitted and received, this is the most fundamental yet necessary component of vocal communication. Within level two, token features are the basic components of vocal communication. Vocal communication is made up of small units of sound; the energy in the signal does not alter meaning, and there exist relationships between objects and sounds. These features fall into level two as they relate to small units. Level three, message features, include features relating to words and vocabulary. This is a step up from small units, as these features describe words and how they are used. Finally, within level four, language features, are the features relating to language expansion and learning. This goes beyond just words and their use, as we are studying everything encompassed in one animal's vocal repertoire. To note: the use of the word 'language' in this highest level of features does not mean an animal possessing one of these features possesses all the lower-level features. Instead, we use the word 'language' at this upper level since we are discussing the existence of not only a meaningful vocal repertoire of an animal but also their ability to learn and expand this repertoire, an advanced signal of linguistic capability. In the past, researchers, primarily biologists and ecologists, have used traditional approaches to study the existence of these linguistic features in various animals. These methods largely focus on the design of experiments that require interaction with the animals and the analysis of small-scale data.

2.1 Level One: Signal Features

2.1.1 Vocal Auditory Channel and Turn-Taking. This first signal feature can be explained as the exchange of communication by means of creating and perceiving audio signals. Heinrich and Marzluff [115] determined that ravens use vocal auditory channels by conducting a study on a group of 20 wild ravens. Results from statistical analysis showed that yelling is used to advertise social status and hunger. Yelling can be suppressed, however, by social superiors. Gelada monkeys [105] use vocalizations to maintain social bonds between group members. Spectrograms of the recordings were analyzed by Avisoft, and a stepwise discriminant function analysis and MANOVAs were used to determine the probability of correctly assigning vocalizations. Similarly, through playback experiments, dogs have been shown to be able to recognize each other from their barks [169]. By overlaying vocal recordings of meerkats' sunning calls, it was made clear that they possess some form of turn-taking [64]. In order to find evidence for turn-taking, researchers can calculate group-wide call rates and overlap rates to determine if the group-wide call rate influenced the overlap rate. Time-shift analysis by moving the background track against the focal call track can also be conducted to demonstrate the lack of natural overlap.

2.1.2 Broadcast Transmission and Directional Reception. This feature refers to broadcasting and localizing sound. In penguins, recognition of calls can only occur while the calls remain within certain frequencies. Once outside of that range, the penguins cannot locate and recognize the individual making the calls [152]. Playback experiments to document the intensity of the animals' responses and statistical analysis to determine the range of frequency shift that could occur were performed to verify. In Chickadees, it was found that large numbers of D notes in calls can be used to help locate the caller [165]. Similarly, Geladas maintain social bonds with vocalizations [105] and frogs use call matching within groups [95], both of which require the deliberate broadcasting of calls to other group members. Among wrens, duets between group members are common, and vocal interaction rules are learned throughout development [215], depending on the broadcasting of calls to other group members and their young. Analysis of the different vocalizations

Table 1. Linguistic Features in Vocal Communication.

Level	Feature	Description
I	Vocal Auditory Channel and Turn-taking	Communication occurs when an animal emits vocalizations and another receives these signals. Turn-taking refers to the act of taking turns when communicating.
	Broadcast Transmission and Directional Reception	Vocalizations can be sent out in all directions but will be localized in space by the receiver.
II	Reference and Displacement	Reference is the relationship between an object or concept and its associated vocalization or word. Displacement is an animal's ability to refer to objects that are remote in time and/or space.
	Specialization	The output of sound is for the purpose of communication, with unintended sounds not altering meaning.
	Arbitrariness	The meaningful message elements (i.e. signal or word) can be arbitrary. They do not need to share characteristics with the meaning they are meant to convey.
	Duality of Patterning	Meaningless units (i.e. phones) combine to create meaningful units (i.e. phonemes, words, and sentences).
III	Discreteness and Syntax	Vocabulary is made of distinct units; and syntax is the way these units are strung together to form words and sentences.
	Recursion	The ability to apply a syntactic rule to the output of the same syntactic rule, allowing for the addition of more context within a sentence.
	Semanticity	Words have meaning and are used to explain features of the world.
	Prevarication	With access to vocal communication, animals also have access to lie and deceive one another.
IV	Openness	The ability to generate new words or create new messages.
	Traditional Transmission	Acquisition of language through teaching and learning with conspecifics.
	Learnability	The ability of an animal to learn the language of another species or an unfamiliar dialect from the same species (if it exists).

produced, dependent on context, can be done manually and with different categorizing systems. The type of reaction from group members to certain vocalizations can be very informative, and these reactions are also manually categorized based on factors such as rate and frequency.

2.2 Level Two: Token Features

2.2.1 Reference and Displacement. Reference, referring to objects in the current vicinity, and displacement, referring to objects that are absent in the current space, have been observed in animals outside of humans. Female baboons adapted their grunts to demonstrate a location change when in different environments [80]. Chickens, an animal with a more rudimentary level of vocal communication, have distinct differences in their calls for a ground predator versus an aerial predator [191]. They also have clear food calls and calls to indicate where said food can be found, suggesting that there are associated sounds with objects that can be communicated to other chickens. Taking it one step further, there is evidence that animals have the ability to communicate about objects outside of their sight, as well as plan for the future. Apes have been found to communicate about absent entities and can even keep track of the location of a desired object [164]. When it comes to threats, female orangutans were tested using predator presentations. They

vocalized only after the predator stimuli were removed, possibly waiting until it was safe to communicate about the threat [143]. Extending this feature further, the laughs of hyenas potentially encode information about the producer's social status and age [170]. Using statistical analysis, researchers can compare different acoustic qualities of calls to understand the patterns of how vocalizations change depending on context. Over time, the consistent relationship between a vocalization and a specific object or situation allows researchers to show evidence for reference. Studies where an object is shown to animals and then removed to see whether they can still communicate about the absent entity have been used to test displacement.

2.2.2 Specialization. Specialization is the idea that the output of sound functions merely as a means of communication, and unintended physiological noises shouldn't alter meaning. In ravens, the rate and types of food calls are referential, but it also appears that increases in call frequency were purposeful [39]. The amount of food affected the calling rate i.e. there was an increase in calling rate once food was present, demonstrating that the calls were purposeful and not random. Dog pants are another example of this language feature, as the panting is not used as a means of communication (a signal), but merely a way to cool off (it is instead a cue) [35, 204]. The length of a vocal message can also have significance. In Mexican free-tailed bats, calls frequently emitted were brief, while the calls emitted during times of distress were notably longer. Analysis of their call syllables revealed that across four species, brevity was applied during normal social calls, and was disregarded in times of stress. The length, proved most important as it was serving as a means of communication [162]. To study this communication feature in animals, the purposes of vocalizations are statistically analyzed in different contexts. Playback experiments that add natural physiological noises would be an effective way to determine the degree to which specialization occurs in animals. Oftentimes, the variation in calls comes from the ordering of notes or the rate of the notes being produced, with very little research to suggest that additional unintentional noise affects the message.

2.2.3 Arbitrariness. Rats can distinguish between sequences of vocalizations regardless of the length of said sequence [62]. A series of experiments exposed rats to stimuli. Alternating pitches were significant, but the length of vocal sequences was not. Among wrens, there is some flexibility in their calling. Altering syntax, rhythm, and spectra still produced a response, as long as a call contains its essential song elements. [121]. Similarly, song elements matter in the vocalizations of Carolina Chick-a-dees. Using playback experiments and ANOVA analysis, researchers found that vocalizations with C-call types caused the birds to approach sites more frequently. Additionally, many c notes in a call could represent novelty [90]. Similarly, it was found that the call length in Chickadees seems unaffected by the closeness of a predator [13]. Stimulus presentations were conducted in a room then the recordings were manually analyzed. As the call length was not clearly affected by the closeness of the predator, results suggest that information about proximity is actually relayed through the rate of calling. Presenting animals with recordings that vary in both pitch and duration allows researchers to test whether the length of a vocalization or call changes its meaning. Similarly, studying the vocalizations produced in response to different stimuli can demonstrate whether or not call length is affected by different situations. Performing statistical analysis on the data collected from animal responses, either positive or negative, allows researchers to test arbitrariness.

2.2.4 Duality of Patterning. Duality of patterning describes two levels of structure: meaningless units (i.e. phones) make up meaningful units (i.e. phonemes and words). This creates a large repertoire of meaningful units from only a small set of meaningless units, preventing the need to create an entirely new vocal unit whenever a new object or concept needs to be referenced. The Japanese great tit songbird has been documented to exhibit this feature through

the use of discrete notes in their call repertoire [247–249]. Studies conducted by Suzuki et al. [249] showed that these birds combine three discrete notes ('A', 'B', and 'C') into the 'ABC' call which is meant to warn conspecifics about predators. They also have a 'D' call to attract conspecifics and combine the two calls into the 'ABC-D' sequence to gather conspecifics to mob a stationary predator. Duality of patterning is also seen in sperm whales. Schulz et al. [230] reported that sperm whales combine clicks, which are not meaningful on their own, in specific patterns to create multiple coda types used to interact with other whales, especially in the same unit or clan, facilitating social relationships.

2.3 Level Three: Message Features

2.3.1 Discreteness and Syntax. Discreteness and syntax are the idea that words are made up of discrete units combined together based on a defined set of rules. Monkeys have been found to assemble individual calls into larger units [9]. Through experiments playing predator sounds to monkeys, researchers noted that putty-nosed monkeys will combine vocalizations into different call sequences linked to external events. In cats, 71 acoustic units were identified by generating spectrograms from a sample of recordings of 13 domestic cats [190]. The sound files were analyzed in Raven Pro, and through visual analysis, researchers organized the vocal sounds hierarchically. Brazilian free-tailed bats have demonstrated that they follow specific syntactical rules [29] and zebra finches can learn to discriminate between two stimulus sets that are only different in the way their elements are ordered [256]. The Madagascar tree frog seemingly can choose single call notes from a pool of notes and assemble them in ways or orders that are not predictable [181]. Recordings can be turned into visual spectrograms, allowing researchers to easily identify different notes within the calls as well as the lengths of specific notes. Through statistical analysis, the arrangement of notes and how often each note is used in different contexts can be studied. If one note is frequently used at the beginning of a call during a food presentation experiment, for example, this helps verify syntactical structure as the animal is using a discrete unit of sound repeatedly over time in a specific context.

2.3.2 Recursion. Recursion is the ability to apply a syntactic rule to the output of the same syntactic rule, allowing for the addition of more context within a sentence. Among starlings, nine out of eleven were able to classify sequences from two grammars. This showed that starlings can identify syntactically well-formed strings that contain recursive elements [94]. Another study on humpback whales found that their songs are learned in segments which are then combined into themes as a whale's knowledge acquisition grows [91]. More recent studies have dealt with this linguistic feature, with Garland's research using a large database and a self-organizing map to derive its results. Since few papers have dealt with this feature of vocal communication, there is no clear consensus on the methodology used to study this idea. To understand whether animals use recursive units of sound within calls, researchers can use playback experiments to determine if animals respond to calls where units have been synthetically added. Additionally, finding statistically significant evidence of units of sound being repeated in an animal's vocalizations helps to test this feature.

2.3.3 Semantics. Semantics is the idea that words have meaning and are used to explain features of the world. Monkeys seem to be able to convey information about external events [24]. A later paper on titi monkeys also supported the idea that they can convey complex knowledge about the external world to listeners [23]. Prairie dogs distinguish between individual predators and incorporate physical information about the predators in their calls [237]. A Unigon real-time analyzer was used to study call elements in these recordings, and 12 variables were identified and then analyzed with discriminant function analysis. Chickadee calls likely contain information about the degree of threat that a predator presents [251]. These birds also have four note types, with information encoded within these calls [82]. Accounting for effects of sex, age, environment, and more, researchers can create several databases with these factors in

mind to study. Predator presentations and playback experiments are commonly used to study the resulting vocalizations and behaviors of animals. If call sequences are frequently linked to specific events, researchers can study this statistically to determine the significance of the relationship. Additionally, the number of a certain note(s) in calls or the rate and frequency of notes within calls is studied to relate vocalizations to specific objects or situations.

2.3.4 Prevarication. This feature describes an animal's ability to lie and deceive another. Drongo birds showed that they vary their alarm calls to maintain deception [87]. The researchers studied whether the number of calls or call types changed during response calls and whether the target birds change their alarm response. Brown thornbills use vocal deception to deter nest predators by producing calls that delay the predators' feeding response [12]. There is some evidence that prevarication exists in chickens. Researchers have been able to show that male chickens alter their calling based on who the receiver is, which can help put them in a better hierarchal position [168]. Tufted capuchin monkeys will use false alarm calls during feeding in order to discourage competitors, a clear use of prevarication in their communication [132]. By setting up experiments with different pairings of animals, researchers can determine if animals alter their vocalizations based on who they are near. Using statistical analysis and plotting, the changes in vocal patterns can be compared with the changes in animal groupings to determine the degree to which vocal deception is used.

2.4 Level Four: Language Features

2.4.1 Openness. The ability to generate new words or new meanings is referred to as openness. By using different acoustic elements, chestnut-crowned babblers will arrange vocalizations to make different sorts of vocalizations [73]. Using recordings, five parameters were extracted from five elements in two of these birds' call types. Humpback whale songs have increased in complexity over time and gradually changed, while simultaneously becoming more individually unique [3]. Analysis was done using the Levenshtein distance similarity index (LSI) to arrive at this conclusion. Japanese monkeys have locale-specific dialects, which require a degree of openness as new signal patterns can be spread through a population within a single generation. This was observed from a large database of recordings taken over a prolonged period of time at multiple sites [102]. Additionally, zebra finches use syllable chunking to create individual songs. By extracting syllables from songs of older group members, they can string syllables together in unique ways [268]. One way this language feature can be studied is evolutionarily. By tracking changes in an animal's language, new songs or ways of arranging notes can be discovered. This can also be studied from a learner's perspective, noting how offspring learn the songs of their parents. The syllable orders and arrangements are compared between parents and their offspring, and the changes can be statistically analyzed to notice what changes occur and how syllables get rearranged.

2.4.2 Traditional Transmission. Traditional transmission (also known as cultural transmission) refers to the animals' ability to pass down their own language. Dolphins can identify one another using their whistles, which would, by implication, need to be learned over time [124]. Songbirds from seven different families had predator calls that remained the same, meaning it is clearly a learned call that is used to keep safe regardless of which family a songbird is from [267]. Songbirds were also found to gradually produce distinct syllables, suggesting that throughout development, language is acquired and then slowly produced [158]. If across multiple families or groups of animals, calls remain the same, there is strong support for the idea that language can be taught and passed down. Studying over time how language either remains the same or changes can be evidence that if language is taught and passed down, it will largely remain the same. Additionally, the expansion of an animal's language over time is further evidence for this feature.

2.4.3 Learnability. Learnability is the ability to learn another species’ language. Currently, there is little research to support animal species’ abilities to learn the languages of other animals. The Japanese tits responded to Willow tit calls similarly to how they respond to their own group members’ calls. By replacing the D call with the *taa* call from Willow tits, researchers found that the Japanese tits responded equally to the ABC-D and ABC-*taa* calls. Responses were only reduced when the ordering of the call syllables was changed [249]. There is also research on the abilities of animals to learn the “language” of other groups of the same species. For instance, there is some evidence of song sharing between the Madagascar and Gabon whale populations [214]. Additionally, tamarins, in an experiment, were able to learn one grammar (the FSG), but not the other (the PSG), that had been created outside their repertoire [86]. To learn the language of another animal would mean learning outside of an animal’s existing vocabulary. Creating artificial forms of language and seeing how much an animal can learn is a way to test for the existence of this language feature.

3 Computational Approaches to Animal Linguistics

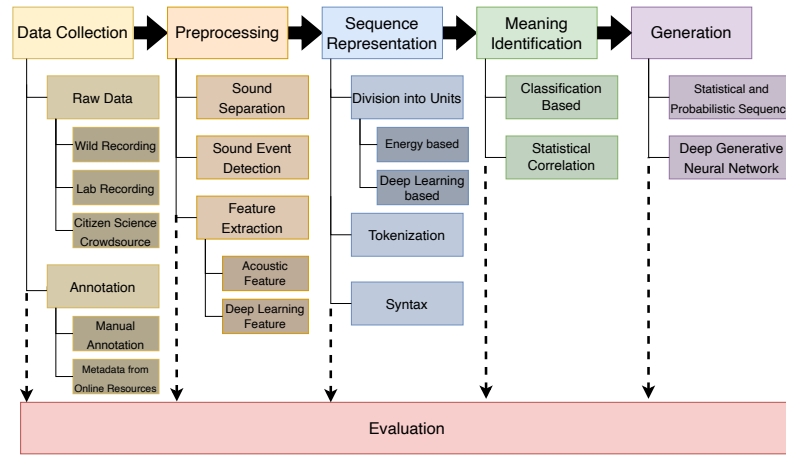


Fig. 4. Existing Concepts and Tasks of Computational Animal Linguistics.

From Section 2, we see that researchers across multiple disciplines have done extensive studies to find evidence for the existence of linguistic features in animals. However, they have relied mostly on human manual work and traditional statistical analysis techniques to carry out experiments. In the past decade, we have seen the significant advancement of data-driven machine learning models. They have proven usage in processing the ever-increasing amounts of data available and automating many tasks at a large scale unsuited for manual work. Researchers have found new ways to apply more computational approaches to enhance their workflow and enable experiments that were virtually impossible to conduct in the past.

We will structure this section based on a 5-step pipeline that generalizes the research process to solve problems in computational animal linguistics (see Figure 4). We first start with the data collection process, where researchers build large datasets of animal vocalization audio recordings and videos. We immediately follow that with the data processing step, where computational methods are applied to separate noisy artifacts and extract clean animal vocals, as well as create features to serve the next steps in the pipeline. Then, we discuss how these animal vocals are further divided into smaller, more fine-grained units and tokenized into symbols to provide a representation and help us discover their

structure. Those representations can be further analyzed using well-designed experiments to confirm associations with semantic and contextual meaning. Lastly, we will discuss methods to artificially generate specific, targeted animal vocalization as a vehicle to attempt playback experiments and potentially, someday in the future, perform translation between human and animal.

3.1 Data Collection

Data collection determines both the ecological validity of subsequent analyses and the supervision available to computational models. We distinguish *recording regime* from *resource type*: recordings may be focal, passive acoustic monitoring (PAM), laboratory-based, crowdsourced, or collected from online media, whereas the resulting resources may be archives, packaged datasets, benchmarks, research initiatives, or private corpora. These categories are not interchangeable, particularly for model development and evaluation [127, 211]. We first review major collection and annotation regimes and then compare representative reusable datasets and benchmarks in Table 2.

3.1.1 Field Recordings: Focal and Passive Acoustic Monitoring. Field recordings broadly follow focal or passive regimes. Focal recording targets a known or visually tracked source, often using directional microphones or hydrophones, which facilitates source attribution and detailed metadata but is difficult to scale. PAM instead uses stationary or autonomous sensors to collect long-duration soundscapes, providing broader ecological coverage at the cost of increased noise, overlapping callers, and uncertain source identity [211, 221]. This distinction also creates a machine-learning domain shift when models trained on relatively clean focal recordings are deployed on PAM soundscapes [109, 127]. Recording hardware depends on environment and taxon. Marine studies use hydrophones, as in the Watkins Marine Mammal Sound DB, Public Orca Dataset, and Project CETI [7, 192, 225]. Terrestrial monitoring uses directional microphones, autonomous recorders, and microphone arrays, including Congo Soundscapes and the Gombe Chimpanzee Archive [227, 270]. Ultrasonic taxa require specialized detectors, as reflected in resources such as ChiroVox and NABat [101, 255].

3.1.2 Laboratory Recordings: High-Fidelity Audio and Rich Metadata. Laboratory settings reduce environmental noise and permit controlled manipulation of vocal production and perception [221]. Mouse ultrasonic vocalizations, for example, are commonly recorded in sound-attenuating chambers [81], while multi-channel microphone arrays can partially address the “who said what” problem by assigning calls to individuals [264]. Controlled studies can also align audio with physiological or anatomical measurements: songbird vocalizations have been paired with in-vivo electrophysiology [50], and primate vocal production has been studied using X-ray video and computational modeling [85]. Such recordings provide high-fidelity signals and rich metadata, although laboratory conditions may not fully reflect natural behavior [221].

3.1.3 Citizen Science / Crowdsourced Recordings. Citizen-science platforms enable bioacoustic collection at scales difficult to achieve through individual field studies. xeno-canto [258] and iNaturalist [48] allow contributors to upload recordings with taxonomic, geographic, and temporal metadata, while eBird observations are linked to media archived in the Macaulay Library [245]. These resources support broad geographic and temporal coverage but vary in recording quality and contributor expertise, requiring validation mechanisms such as community review and filters for unusual observations [33]. Citizen-science systems may also integrate automated analysis, as in BirdNET [130], or support targeted monitoring efforts such as the New Forest cicada project [282] and Soundscapes to Landscapes [54].

3.1.4 Leveraging Online Videos for Data Collection. Online video platforms provide another source of large-scale, naturally occurring animal vocalizations. YouTube and TikTok have been used to construct canine corpora capturing

“in-the-wild” behavior, including DogSpeak, the Canine Age Transition Vocalization Dataset, and Dog2Vec [150, 151, 156]. Because online retrieval returns both irrelevant videos and acoustically noisy recordings, these pipelines combine several stages of automated filtering and preprocessing. DogSpeak and the longitudinal canine dataset first filter candidate videos using visual information from thumbnails together with textual metadata such as titles, descriptions, and comments [150, 151]. The resulting audio is further cleaned using language-guided source separation and sound event detection to isolate canine vocalizations from background sounds and non-vocal segments [150, 151, 156]. More broadly, AudioSet demonstrates the scale of online audiovisual data, with millions of YouTube clips spanning domestic, livestock, and wild animal sound classes [93].

3.1.5 Annotating Metadata from Online Sources. Online sources can also provide contextual metadata unavailable from audio alone. DogSpeak verifies individual identity, sex, and breed from source-video information [151], while longitudinal canine data extend this process to birthdate and developmental age [150]. EmotionalCanines derives arousal and valence labels from observed video context [60]. Annotation granularity varies substantially across datasets: iNaturalist Sounds generally provides weak recording-level taxonomic labels [48], whereas ENABIRDS contains strong time–frequency annotations [53]. HumBugDB additionally provides metadata such as mosquito species, age, sex, and capture method [137]. Tools such as Praat, Raven, and Praat++ support manual acoustic inspection and time-aligned annotation [27, 126, 280], while ANIMAL-SPOT and OpenSoundscape support automated detection and classification at larger scale [19, 145].

3.1.6 Representative Public Datasets and Benchmarks. Because the resources above serve different purposes, Table 2 is representative rather than exhaustive. We include reusable resources that are publicly available or accessible through a documented process, provide sufficient information for comparison across recording regime, annotation type, task, scale, and access terms, and represent distinct collection or evaluation settings. We also include established benchmarks that standardize evaluation across datasets or recording conditions. Private training corpora, research initiatives, software platforms, and general-purpose archives are therefore discussed in the text rather than treated as equivalent to packaged datasets or benchmarks. Reported scale is preserved in the unit provided by each source rather than converted to estimated recording hours. The avian benchmarks particularly expose the focal-to-PAM gap. BirdSet combines large-scale focal training data with eight strongly labeled PAM evaluation sets [211]; BirdCLEF evaluates species identification using focal training recordings and hidden soundscapes [127–129]; and BIRB evaluates retrieval and generalization from focal recordings to passive recordings [109]. Thus, BirdSet provides standardized multi-dataset evaluation, BirdCLEF a recurring challenge setting, and BIRB an explicit test of robustness under recording-domain shift.

3.2 Data Pre-Processing

This section details the transformation of raw recordings into clean vocal segments and features for the computational pipeline.

3.2.1 Sound Separation. Computational analysis requires pre-processing to isolate target signals, typically categorized into the following sub-tasks. **Denoising** suppresses environmental noise (wind, insects, speech) while preserving focal vocalizations. A standard benchmark is *noisereduce* [222], which employs spectral gating. To improve performance without clean training data, Miron et al. [171] utilized speech enhancement models (Demucs [63], CleanUNet [140]) to generate pseudo-clean targets, retraining on synthetic mixtures with time-scaling to accommodate species-specific pitch

Table 2. Representative reusable bioacoustic datasets and benchmarks.

Resource	Taxa / scale	Setting	Labels / task	Access / terms
BEANS [107]	Multi-taxa; 12 public datasets	Mixed	Dataset-dependent labels; classification and detection	Public; constituent-dataset terms
BirdCLEF 2022–2024 [127–129]	Birds; annual regional tasks; 2024: >25,000 recordings, 182 species	Focal train + PAM test	Weak focal labels and soundscape targets; species identification	Kaggle; edition-specific terms
BirdSet [211]	Birds; >6,800 h training, >400 h evaluation	Focal train + PAM evaluation	Weak focal and strong multi-label PAM labels; classification and domain-shift evaluation	Non-commercial; source-specific CC terms
BIRB [109]	Birds; focal corpus + 6 PAM datasets	Focal → PAM	Species presence; retrieval and distribution-shift generalization	Public; constituent-dataset terms
Canine Age Transition [150]	Dogs; 79,142 units / 55,718 sequences (11.4 h), 125 dogs, 6 breeds	Online, longitudinal	Breed, age, age group, and unit transcripts; longitudinal vocal analysis	CC BY-NC-SA 4.0
CatMeows [186]	Cats; 440 sounds, 21 cats, 2 breeds	Controlled contexts	Context, individual, breed, and sex; context classification	Open access; Zenodo states research/non-commercial use
ChiroVox [101]	Bats; 3,902 recordings, 192 species	Focal/reference	Species, specimen, method, and habitat; reference and species identification	Open/consensual; per-record CC terms
DogSpeak [151]	Dogs; 77,202 sequences (33.162 h), 156 dogs, 5 breeds	Online, in-the-wild	Individual, sex, and breed; corresponding classification tasks	CC BY-NC-SA 4.0
HumBugDB [137]	Mosquitoes; 20 h, 36 species or species complexes	Laboratory + field	Time-localized events and species metadata; detection and classification	CC BY 4.0
iNaturalist Sounds [48]	Multi-taxa; 230,000 recordings, >5,500 species	Citizen science	Weak species labels and observation metadata; species classification	Non-commercial research/education; no redistribution
Public Orca Dataset [192]	Killer whale; >225,000 annotations, 23 locations	PAM + focal	Time–frequency boxes and ecotype; detection and ecotype classification	Public (NCEI)

and tempo. **Sound source separation** isolates every signal source (animal, human, environment) into distinct outputs, facilitating the extraction of contextual cues from secondary signals. Key models applied to animal bioacoustics include *MixIT* [65, 269] and *AudioSep* [159, 195]. The **cocktail party problem** involves isolating individual vocalizations from overlapping multi-animal mixtures, a critical step for studying turn-taking. Zhang et al. [277] applied Bi-LSTMs to bat echolocation, while Izadi et al. [122] used U-Net architectures on spectrograms. More recently, *BioCPPNet* [20], a U-Net-based model, demonstrated success across macaques, dolphins, and bats. Additionally, *MixIT* has been adapted for bird song separation [65].

3.2.2 Sound Event Detection. In order to process and analyze animal vocalization at scale when a large portion of the audio recordings may contain non-animal sounds, researchers require automated methods to accurately identify segments that only contain animal vocalization. This is defined as the bioacoustics sound event detection (SED) task. These segments can be more “coarse-grained” such as continuous calls separated by pauses in between, or they can be “fine-grained” individual calls in the same continuous sequence identified by unique call types. In the past, researchers required manual labor to go through each audio recording using software tools to annotate the start and end times

of each segment or vocal unit. Modern computational methods are developed to automate this task and enable the processing and analysis of much larger datasets. These methods are also tailored to each species depending on the characteristics of their calls.

As reviewed by Xie et al. [272], bird calls have been segmented with **energy-based** methods using audio features such as energy thresholds or silence [57, 203, 252, 279, 281] to separate high- and low-energy frames. Denton et al. [65] attempted to perform SED by selecting high-energy 6-second windows likely containing the labeled species, using log mel-spectrograms and wavelet peak detection.

More recently, **deep learning-based** methods have been employed to segment bird calls and classify their type [180, 198, 244, 283] by training vision-based neural network models on spectrogram data. Wang et al. [260, 261] finetuned SED models such as *PANNs* [139] and *BEATs* [49] using a small set of annotated dog barks to perform SED for dog vocalization. Hoffman et al. [120] trained a transformer-based model on 8.8k hours of synthetic data covering a diverse variety of species (e.g., Hainan gibbons, humpback whales, marmosets).

3.2.3 Feature Extraction. A key step for essentially every computational method in animal vocalization is to extract the necessary features from the processed audio data. Features can be defined in many ways depending on which task they serve. They can be **acoustic features** such as energy and silence used to detect animal vocals, or **deep learning based features** such as latent representation vectors used to cluster animal vocals into unique symbols.

There have been many studies dedicated to learning information about individual characteristics, breed, sex, age, context, and emotion in dog vocalization. These studies utilize the Mudi [111, 146, 174, 200] and Mescalina [100, 201] dog bark datasets. To support supervised learning classification tasks, they extract **acoustic features** such as Mel Frequency Cepstral Coefficients (MFCCs), GeMAPS, and eGeMAPS, or a list of chosen features (e.g., pitch, formants, intensity). In other animal species, Sainburg et al. [221] were successful in extracting latent features from dimensionality reduction techniques.

The availability of big data in bioacoustics presents opportunities for new discoveries and applications through the use of **deep learning models** [30], where we train (or pre-train) models, usually through supervised learning or self-supervised learning approaches, to create generalizable representations that can be fine-tuned for a wide variety of downstream tasks (e.g. classification, identification, detection, clustering) [231]. These models are trained using large amounts of bioacoustics data that could cover a wide range of taxa (*AVES* [106], *NatureLM-audio* [216], *Perch 2.0* [257], *MetaPerch* [47], *animal2vec* [226]) or focused on a particular group of species (*BirdNET* [130], *Bird-MAE* [210], *dog2vec* [156]). Several benchmarks have been established to test the performance of various SOTA bioacoustic representation learning models [107, 109, 172, 211]. The validation and test datasets from these benchmarks or the individual studies consist of a variety of sources from different locations on Earth and recording conditions, ranging from smaller-scale focal recordings of targeted species to large-scale PAM recordings covering a long period of time and an unpredictable range of species. These benchmarks provide insightful analyses on the strengths and weaknesses of these models as well as what data, training recipes, deployment conditions, etc. that would optimize performance for their particular use cases.

3.3 Sequence Representation

In this section, we describe how segments of animal vocalization are divided into smaller, finer-grained units in order to create a repertoire of tokens or symbols that represent unique vocal units. Depending on the study, task, or focal species, researchers can define the term ‘token’ in different ways (i.e. clicks and codas in sperm whales, notes in bird

songs, calls and call types in mammals, etc.); some of the terminology in this survey paper are taken from the cited studies. We also discuss how researchers have attempted to analyze the patterns of these representations in order to discover syntactical structure in animal vocalization.

3.3.1 Division into Units. In order to conduct further analysis of vocalizations, longer vocal sequences need to be further segmented into smaller units. There are generally two approaches: **acoustic-based** and **machine learning based** algorithms. Traditionally, **acoustic-based** algorithms offer a solution that is less computationally-demanding by relying on certain thresholds in the acoustic data, which helps define boundaries of the units. Du and Troyer [69] developed an algorithm to automatically segment the notes in zebra finch bird songs by detecting a sudden change in the amplitude envelope. To create a list of candidate vocal patterns that may be associated with specific contexts, Wang et al. [261] applied an energy-based method to segment sequences of dog barks into smaller vocal units. More recently, **machine learning-based** techniques have enabled more accurate solutions that overcome certain technical challenges previously seen with acoustic-based algorithms. For example, to address challenges caused by environmental noise and overlapping bird calls in segmenting bird song recordings, Neal et al. [182] trained a random forest model on spectrogram data to discriminate between bird sound and noise. Also working with bird songs, Ghaffari Jadidi and Devos [97] trained a semi-supervised model using the Mean Teacher algorithm [250] with minimal human intervention in the annotation process; they also provided a comparison between their method and baseline thresholding algorithms [184] to showcase better performance.

3.3.2 Tokenization. For many studies that attempt to analyze the syntactical structure and semantic meaning of animal vocalization, a tokenization step is necessary to categorize individual vocal units into discrete tokens (i.e., symbolic or textual representation), or to transcribe a sequence of vocal units into a sequence of discrete tokens. For example, sequences of sperm whale clicks are manually divided based on pauses and annotated into coda types [21, 234]. More recently, a common approach to automatically perform this task is by the use of **clustering techniques**, where vocal units are clustered into distinct groups based on extracted features [243]. Hagiwara et al. [108] created the ISPA, a system for transcribing animal vocalizations into symbols, aiming for precision, conciseness, and interpretability. They clustered audio input based on acoustic features and feature representations from the pre-trained AVES model [106]. The audio transcriptions were then evaluated on various classification datasets to assess their utility in downstream tasks, showing performance comparable to baseline ML methods.

Wang et al. [260] tokenized dog vocal units into what they defined and referred to as “phones” for dogs, which are minimal sound units analogous to phonetic elements in human language. The authors pre-trained a self-supervised model called Dog-HuBERT (based on HuBERT) on over 20 hours of dog bark sequences. Features from the 11th transformer layer of the model are then clustered using K-Means into 140 clusters, which represent the phones. Similarly, Wang et al. [261] extended this work by training a HuBERT model on 152 hours of dog bark sequences. Clustering was performed through a similar process, resulting in a list of 250 initial vocal units, where each unit is represented by a cluster center (average of frame embeddings) and assigned a unique symbol (integer). These initial vocal units are refined iteratively into 105 “cognitive vocal units” (which they described to be analogous to phonemes in humans) by merging similar ones and removing noise or false units. Each dog bark sequence is thus transcribed into a sequence of these symbols.

3.3.3 Syntax. Once we have transcribed the animal vocals into a dictionary of discrete tokens or symbols (either through the use of manual or automatic methods), we can apply linguistics-based methods to study and model the syntactical structure of animal vocalization. Each animal species can have unique characteristics that define their

structure, such as tokens, timing, repetition, dependencies, etc. In response, researchers need to apply the appropriate methods to capture their structure; some methods may produce good results for some species but fail to model others.

Both Jin and Kozhevnikov [125] and Katahira et al. [131] aimed to discover the model that would best describe the syntactical structure of songs from the Bengalese finch, arriving in some form of a **Markov model**. Jin and Kozhevnikov [125] proposed a modified Markov model, which they termed a partially observable Markov model with adaptation (POMMA), to describe the Bengalese finch song. Their modifications were made to address specific characteristics of these bird songs (i.e., syllable repetition and many-to-one mappings from states to syllables), which they argued could not be captured by a traditional Markov model. Katahira et al. [131] did an independent study in the same year and concluded that the first-order hidden Markov model was sufficient to describe the Bengalese finch song. Kershenbaum et al. [134] conducted a further examination of Bengalese finch songs (as well as other animals) and argued that a non-Markovian Renewal Process can better describe these songs than Markov chains, offering an opposing view to the previous two works and suggesting that further examination is required. Sainburg et al. [220] came to a similar conclusion after analyzing long-distance dependencies of songs from the Bengalese finch (and three other bird species), and proposed that further investigation needed to be done on **hierarchical processes** to account for the non-Markovian patterns that appear in bird songs.

A common approach in Linguistics to define and analyze hierarchical structure is through the use of **context-free grammar**, which builds tree-like structures as opposed to linear sequences, capable of capturing recursion and embedding. For long, the use of context-free grammar was not linked to any non-human species. This changed when works from Gentner et al. [94] and Abe and Watanabe [1] proposed that songbirds have the ability to recognize acoustic patterns defined by context-free grammar based on the results of playback experiments, which could indicate that these songbirds also possess the capability of processing hierarchical structures. In contrast, Van Heijningen et al. [256] and Beckers et al. [15] provided an alternative explanation, attributing this ability to discriminate patterns to acoustic similarity matching as opposed to the acquisition of context-free grammar rules. Another case study for hierarchical structure in animal vocalization is the songs of humpback whales. Payne and McVay [196] provided foundational work to describe the hierarchical structure in humpback whale songs, which inspired numerous works that expanded our understanding of their structure [52, 110, 133, 246]. However, there is still little research done to explore how data-driven computational methods could empower more discoveries for the syntactical structures of humpback whale songs.

3.4 Meaning Identification

Recent progress in animal linguistic processing increasingly focuses on methods for uncovering semantic structure in animal communication. Reviewing this work requires a distinction that is easy to lose in reported performance figures. A classifier that recovers a label from a recording demonstrates that the corresponding information is *present in the acoustic signal and recoverable by an external observer*. Meaning, in the sense used in the study of communication, additionally requires that the information is *produced in a context-dependent manner and interpreted by conspecific receivers* in a way that affects their behavior. These are different claims, supported by different experimental designs, and no amount of classification accuracy converts the first into the second. This section therefore groups representative studies by the type of evidence they provide, rather than by the algorithm they employ.

3.4.1 Types of Evidence for Semantic Claims. We distinguish four types of evidence for interpreting animal vocalizations, summarized in Table 3: source information, production context, receiver discrimination, and referential or label-like function. Source-information and production-context evidence are **observer-side**: they establish what an analyst,

Table 3. Types of evidence for semantic claims about animal vocalizations. Source-information and production-context evidence are observer-side and may be established by classification or statistical association; receiver-discrimination and referential evidence are animal-side and require receiver behavior. Reported classification accuracy, however high, cannot by itself establish an animal-side claim.

Evidence type	Typical task	Example studies	Warrants the claim that...	Does not warrant the claim that...
Source information	Individual identity, sex, age, body size, population or dialect	[45, 96, 102, 130, 135, 146, 149, 157, 170, 179, 183, 186, 205]	the signal carries recoverable information about its producer	this information is encoded as a message; identity cues may be an unavoidable by-product of vocal-tract morphology
Production context	Affective valence, behavioral state, predator class, social setting	[28, 32, 37, 59, 92, 111, 148, 178, 187, 200, 206, 251, 262]	call form co-varies systematically with the circumstances of production	receivers extract the same category, or that the signal is <i>about</i> that context
Receiver discrimination	Playback with differential behavioral response	[1, 14, 42, 46, 90, 99, 104, 152, 169, 176, 238, 253, 266]	the acoustic contrast is perceptible to conspecifics and behaviorally consequential	what the signal denotes; distinguishing two calls is not interpreting them
Referential or label-like function	Playback with referent, addressee, or element-order controls	[9, 10, 23, 55, 74, 78, 188, 189, 194, 219, 237, 248, 284]	the signal functions referentially, as a label, or compositionally	that the underlying representation resembles human lexical semantics

given a recording, can recover from the signal. Receiver-discrimination and referential evidence are **animal-side**: they establish what a conspecific receiver does with the signal. Computational classification alone provides observer-side evidence; animal-side claims require behavioral data from receivers and cannot be established merely by improving a model.

This distinction also clarifies a recurring confound. A model may classify production context with high accuracy because the recording contains information about the caller, the recording session, the location, or the background soundscape, none of which is part of the communicative signal. Production-context results are therefore interpretable as evidence about communication only to the extent that the evaluation controls for these factors, through caller-disjoint and session-disjoint data splits, label-permutation baselines, and ablations that remove or substitute background.

3.4.2 Machine Learning Classification of Meaning and Context. Animal vocalization classification tasks use varied label schemes depending on the species and experimental design, and these schemes provide different types of evidence. **Individual identity** (caller recognition [205]) and **regional dialects or caller-specific features** [130] provide source-information evidence: success shows that the signal is individuated, not that individuation is communicated. **Emotional valence** (e.g., positive vs. negative affect in ungulates [148]) and **behavioral context** (e.g., feeding vs. travel calls in whales [32]) provide production-context evidence because they relate acoustic form to the circumstances of production. **Vocalization type** labels, such as dolphin whistles versus echolocation clicks [67] or elephant rumbles versus roars [66], combine acoustic categorization with an assumed functional distinction: the labels are defined by acoustic form, but their functional interpretation is imported by the analyst rather than demonstrated by the classifier.

Researchers have applied a wide range of computational models to these labeling tasks. Tree-based and ensemble methods (e.g., XGBoost) have been used to predict emotional valence [148], while convolutional neural networks (CNNs) dominate recent bioacoustic pipelines, such as ResNet-based models for orca behavioral classification [32], lightweight CNNs for elephant call-type classification [66], and spectrogram-based CNNs for dolphin calls [67]. Recurrent neural networks (RNNs) and long short-term memory (LSTM) models are applied to sequential vocal data, such as marmoset call patterns [279]. Classical classifiers—SVMs, GMMs, and random forests—remain widely used baselines [174]. In comparative analyses, CNN-learned embeddings with data augmentation consistently outperform handcrafted acoustic features [2, 130].

Interpretability methods bear directly on the production-context confound because they expose *what* a model used to reach its decision. Prototype-based architectures such as AudioProtoPNet [116] attribute a prediction to specific spectro-temporal regions of the input, making it possible to check whether a model attends to the vocalization itself or to correlated background structure. Such evidence is not a substitute for receiver behavior, but it materially strengthens a production-context claim in a way that headline accuracy cannot.

Evaluation metrics vary across tasks but most commonly include **accuracy**, **F1-score**, and **area under the curve (AUC)**. For example, CNN models for dolphin calls achieved up to 95.2% accuracy ($F1 \approx 0.88$) [67], while orca behavior classification reached 96.1% [32]. Studies with class imbalance often report balanced accuracy or unweighted average recall (UAR) [148]. Binary affective classifiers report high AUCs. In multi-class settings, macro-F1 and precision–recall curves are used to assess per-class performance, while detection tasks report frame-level or segment-level F1 [279]. Across species, reported accuracies frequently exceed 90%. These figures establish that the labeled distinction is recoverable from the acoustic signal under the given recording conditions; they do not establish that the distinction is one the animals themselves encode or interpret. Reported performance is best read as an upper bound on decodability, and comparison across studies is further limited by the fact that datasets differ in the number of callers, the balance of classes, and whether test splits are disjoint by individual.

3.4.3 Statistical Correlations Between Calls and Context. A complementary strategy for investigating semanticity dispenses with complex classifiers and asks whether call types co-vary with context in statistically reliable ways. Wang et al. [261, 262] mined large-scale online videos of Shiba Inu dogs and estimated conditional probabilities $P(\text{context} \mid \text{call})$ and related association measures. They found that growls were disproportionately associated with agonistic or interactive contexts, while whimpers divided into distinct subtypes correlating with either attention-seeking or discomfort. A similar correlation-based approach was used by Prat et al. [205] in extensive recordings of Egyptian fruit bats: by annotating the social context of thousands of interactions and linking these with acoustic measurements, and validating with balanced-accuracy matrices and permutation tests, they demonstrated that vocalizations reliably predicted both the nature of the interaction and its likely outcome. Finally, Andreas et al. [7] outlined a roadmap for sperm whale research within the CETI project, emphasizing large-scale statistical modeling and unsupervised learning to discover correlations between codas and social or ecological variables.

Because permutation tests and disjoint splits guard against precisely the leakage described above, well-controlled association studies furnish stronger production-context evidence than accuracy figures alone. They remain observer-side evidence nonetheless: a stable form–context regularity is a property of production and establishes the existence of candidate form–meaning pairings rather than their communicative interpretation.

3.4.4 Playback Experiments Testing Receiver Interpretation. Establishing animal-side evidence requires manipulating the signal and observing receivers. In controlled playback experiments, synthetic or natural calls are presented to

conspecifics and their behavioral responses are measured; if animals respond in a manner consistent with a hypothesized meaning, the evidence bears on interpretation rather than production alone. Suzuki et al. [248] applied this design to compositional syntax in Japanese great tits. By splicing recorded call elements into novel sequences, they showed that birds responded to an “ABC-D” sequence with both vigilance and approach behaviors, whereas the reversed sequence elicited no such response. This provides referential or compositional evidence because the manipulated variable is element order and the response tracks it.

Not every playback result provides referential evidence. A substantial body of work establishes only that receivers *discriminate* between call variants. Freeberg and Lucas [90] showed that Carolina chickadees respond differently to chick-a-dee calls varying in note composition, and Campbell et al. [42] that black-capped chickadees distinguish natural from manipulated calls; in domestic dogs, Molnár et al. [176] found that both context and caller identity affect discrimination between barks. These designs provide receiver-discrimination evidence by showing that an acoustic contrast is perceptible and behaviorally relevant, without determining what, if anything, the contrast denotes—a distinction that matters when such results are cited in support of semantic claims.

Playback designs have also been applied to large mammals. In elephants, Pardo et al. [194] confirmed machine learning predictions by playing rumbles originally directed at particular individuals; elephants reacted more strongly to rumbles addressed to them than to those addressed to others, indicating that they recognized when a call was intended for them. Similarly, Oren et al. [188] used playback of marmoset “phee” calls and found that individuals selectively responded to calls produced for them, with above-chance receiver decoding, shuffled baselines, and family-level similarity in the discovered labels together arguing for shared, learned vocal labels. These two studies illustrate a productive pattern for the field: a classifier generates a candidate form–meaning hypothesis from source-information or production-context evidence, and a playback experiment tests that hypothesis using receiver-discrimination or referential evidence. The roadmap of Andreas et al. [7] makes the same recommendation explicit, proposing interactive playback as the gold-standard evaluation for competing semantic hypotheses in whale communication.

3.5 Generation

Computational work on generating animal vocalizations remains limited, but two main approaches have emerged: statistical sequence models that replicate structural rules in vocal repertoires, and deep generative networks that synthesize realistic waveforms or spectrograms for analysis and augmentation.

3.5.1 Statistical and Probabilistic Sequence Models. Probabilistic models aim to capture the sequential organization of animal calls in a transparent, hypothesis-driven way. Lu et al. [160] used Partially Observable Markov Models (POMMs) to model Bengalese finch songs, showing that hidden states govern context-dependent syllable transitions. Their generative sequences reproduced key *n*-gram statistics of natural songs, while comparisons with simpler Markov assumptions revealed long-range context effects. After deafening, song structure collapsed to simpler transitions, demonstrating that auditory feedback maintains syntactic organization. These models thus provide interpretable tools for testing rules of vocal sequence production.

3.5.2 Deep Generative Neural Networks. Neural approaches focus on producing calls through learned latent representations. Reilly et al. [213] introduced BiWaveGAN for mouse ultrasonic vocalizations, generating waveform-level calls that elicited naturalistic behavioral and neural responses during playback. This receiver-side validation provides evidence that the generated calls preserve biologically relevant properties. Similarly, Guei et al. [103] treated bird spectrograms as images and used a GAN-like system to generate synthetic songs; when used for augmentation, these

samples improved species-classification accuracy by over ten percent. This result demonstrates computational utility for data augmentation, but does not by itself establish that the generated songs are perceived as natural or communicatively meaningful by birds. More recently, Paradise et al. [193] introduced WhAM, a Transformer-based generative model fine-tuned on approximately 10,000 sperm whale coda recordings. WhAM generates synthetic codas from audio prompts and supports acoustic translation into coda-like signals. Its outputs were evaluated using Fréchet Audio Distance and perceptual judgments by expert marine biologists, while its learned representations were assessed on rhythm, social-unit, and vowel-classification tasks. These evaluations provide evidence of acoustic fidelity, expert-perceived plausibility, and computational utility, but do not yet constitute receiver-side validation by sperm whales. Generative models can therefore supply scalable training data, while their suitability as biologically plausible experimental stimuli depends on receiver-side validation.

Evaluation of generated animal vocalizations should distinguish acoustic fidelity, computational utility, and biological validity. Sequence statistics and acoustic-distribution measures can establish whether generated outputs reproduce selected properties of recorded calls, while improvements in downstream classification demonstrate their usefulness for data augmentation. Neither form of evidence alone establishes that an animal would perceive a generated signal as natural or communicatively meaningful, because synthetic samples may contain biologically irrelevant artifacts or task-specific cues. Strong claims of biological plausibility therefore require receiver-side validation, ideally through controlled playback experiments measuring behavioral, physiological, or neural responses. Under this distinction, ECOGEN [103] provides evidence of augmentation utility, whereas BiWaveGAN offers stronger evidence of biological relevance through playback-based validation.

4 Significant Case Studies in Animal Communication

This section uses methods from Sec. 3 to three taxonomic groups to better understand what computational analysis has been done for each. We choose songbirds, marine mammals, and domestic dogs for three reasons: these groups are the most studied across both biological and computational literature (Figure 7, Sec. 7), have large corpora for researchers to analyze, and largely use different methods to collect and measure data—songbirds in controlled lab environments or field detectors, cetaceans with passive acoustic monitoring, and dogs with close range video recordings. This diversity in data fosters a diversity in downstream research questions: syntax and scalable species identification in birds, culturally transmitted combinatorial structure in cetaceans, and species and unit discovery in dogs. These case studies are illustrative rather than exhaustive, and we focus on groups for which computational work has produced especially clear empirical results.

4.1 Case Study 1: Songbirds

Birdsong remains a central model for learned vocal communication and has increasingly been studied with automated computational methods including supervised, self-supervised, sequence-modeling, and generative methods [272]. Birdsong is primarily available in two formats: individual recordings in isolated lab environments—alongside an expert-annotated subset of syllable labels—and large-scale community recordings in the field without any such unit-level annotations (Sec. 3.1). The former data type suits syntactic work, and the latter suits monitoring work; thus, these data types overlap minimally in both collection and subsequent analysis [272].

TweetyNet applied CNNs to continuous birdsong and achieved expert-level syllable segmentation across species [56]. Later, *TweetyBERT* used self-supervised methods to learn species-invariant embeddings without labels, reducing annotation costs for large-scale analysis [259]. Recurrent language models kept predictive information from sequential

elements farther back than the immediately preceding one or two elements [178], and the Transformer *FinchGPT* reproduced this with attention concentrated on distant syllables [138]. In transfer learning for biodiversity monitoring, large-scale field recordings have benefited from pretrained audio encoders such as *BirdNET*, *PaSST*, and *EfficientNet* variants, which improve generalization to unseen habitats [98]. Models such as *MFF-ScSEnet* and *DuSAFNet* use attention or fusion variants to add modest generalization gains [155, 161]. Generative approaches such as *ECOGEN* and diffusion-based frameworks create realistic bird vocalizations from learned latent codes for augmentation and related analysis [103], but have been evaluated by acoustic similarity instead of playback to birds. These models also provide computational accounts of how song patterns can be produced from learned representations [31, 235].

4.2 Case Study 2: Marine Mammals

Marine mammal research has benefited from large passive acoustic corpora and sequence-level analysis of codas, calls, and song. This includes noisy, sparsely labeled long-term hydrophone archives: sperm whale codas from a Dominica clan [7], eight years of humpback song [11], and roughly 19,000 hours of killer whale audio [192].

Recent work on sperm whales identified two context-sensitive features, *rubato* and *ornamentation*, which combine with context-independent *rhythm* and *tempo* to form a sizable combinatorial “phonetic alphabet” [234]. In humpback whales, infant-speech-style segmentation applied to eight years of recordings uncovered statistical structure, including Zipf-like distributions and coherent subsequences [11]. Together, these studies strengthen the case for culturally learned and rule-governed sequencing in great whales. As work becomes more specialized towards dialectal and repertoire analysis, existing data archives have become tractable through automated detection, separation, classification, and localization. *ORCA-SPOT* found call segments from the Orchiade recordings [18]; *ORCA-SPY* added classification and localization so that pod and year are attached to each call [113]. *ORCA-WHISPER* then extended these works by using GANs to synthesize realistic call-type exemplars for augmentation and hypothesis testing [17]. CNNs and object detectors also processed passive acoustic recordings of seals into ecological indicators like seasonal call counts that are linked to sea ice and bathymetry [75–77, 239].

4.3 Case Study 3: Domestic Dogs

Canine vocalizations have relatively smaller but higher-quality corpora compared with those of songbirds and marine mammals. The single-breed Mudi database containing 6,646 barks from 14 dogs [174] allowed for identification of individual identity, sex, age, and context. Multi-label datasets, such as *DogSpeak* [151] and *EmotionalCanines* [60], and other unlabeled canine recordings [156] can also be used for recognition and unit discovery.

Simpler methods consisting of manually created features layered with shallow classifiers found that barks can reliably encode caller information. These methods outperformed humans in identifying individuals and predicted age and sex of dogs with high accuracy, 80% and 85%, respectively [146]. Additional supervised methods correctly identified positive and negative barks with F-measures above 0.87, with arousal and valence at $r \approx 0.63$ – 0.70 [202]. However, poor generalization across multi-breed data [151] and finer-grained tasks like emotional state detection fostered the use of learned representations and unsupervised methods [60]. Human-speech encoders such as *Wav2Vec2* beat feature baselines [2]. *Dog2vec*, a dog-specific bioacoustic embedding model, was pretrained on 6,000 hours of bark audio and also outperformed general bioacoustic encoders [156]. Unit discovery using minimal-pair algorithms has found recurring sequences of phone-like units, with some tied to activities [261]. HuBERT clustering also finds phone sequences whose n-grams correspond with canine behavior [260].

5 Potential Applications

The computational pipeline in Sec. 3 transforms raw vocalizations into structured representations, enabling scalable, objective, and non-invasive applications across scientific and applied domains. Below, we highlight key areas where these capabilities are already impactful or rapidly emerging.

Human–Animal Interaction and Affective Communication. Computational models can infer affect, context, and identity from companion-animal vocalizations, improving communication and welfare. Supervised and transfer-learning approaches classify emotional valence and behavioral context from dog barks and cat meows using acoustic and deep representations [2, 60, 111, 175, 187]. Behavioral evidence further shows sensitivity to human vocal cues and cross-species emotional contagion [36, 144, 208, 212]. These results support emotion-aware interfaces and adaptive training or enrichment systems.

Veterinary Health and Precision Livestock. Bioacoustic analysis enables continuous, non-invasive monitoring of health and welfare in livestock. Machine-learning models detect respiratory disease, stress, and distress in cows, pigs, and poultry from vocal features [88, 123, 199, 217, 224]. Vocal signals also encode social and physiological information and can be used for calming interventions [36, 153, 275]. These methods align with precision-livestock systems by scaling assessment while reducing labor and observer bias.

Conservation and Acoustic Monitoring at Scale. Passive acoustic monitoring (PAM) provides continuous, low-impact sensing of biodiversity. Advances in denoising and sound-event detection improve segmentation in noisy environments, enabling reliable downstream inference [120, 171]. At scale, automated classification supports species identification, behavior analysis, and ecosystem monitoring from long-duration recordings [40, 242, 270]. These systems effectively convert soundscapes into persistent ecological sensors.

Environmental and Climate Indicators. Animal vocalizations themselves can act as indirect indicators of environmental change, predator presence, or anthropogenic intrusion, providing complementary signals to traditional sensing modalities [26, 41, 71, 197]. In marine contexts, PAM is routinely used for tracking vessel activity and ecosystem dynamics alongside marine mammal vocalizations [117]. These capabilities suggest a broader role for computational animal linguistics in low-power, distributed, and covert environmental intelligence systems. At the same time, such applications raise important ethical and governance considerations regarding dual-use technologies and unintended surveillance. In birds, temperature and habitat shifts alter song structure and responsiveness [58, 173, 274]. Marine mammal vocalizations similarly track oceanographic variation [89]. Combined with traditional ecological knowledge [4], these findings position bioacoustics as a complementary tool for monitoring climate impacts.

Bionics, Robotics, and Communication Technologies. Animal-inspired vocal systems are advancing robotics and communication. Biomimetic synthesizers enable naturalistic robot vocalizations [177], while marine-mammal-inspired signaling supports underwater communication [44, 154]. Generative models enable controllable call synthesis for analysis and playback [68, 278]. Emerging foundation models and embedded systems further enable generalizable representations and on-device inference [106, 229].

Scope and Outlook. Across domains, a unified pipeline—collection, preprocessing, representation learning, structure discovery, inference, and synthesis—enables scalable and minimally invasive analysis of animal communication. Existing applications in welfare monitoring, conservation, and interaction demonstrate feasibility. Key next steps include

improved validation with behavioral and physiological ground truth, expansion of foundation models to non-human acoustics, and integration into multimodal decision systems for scientific, industrial, and policy applications.

6 Challenges in Computational Animal Linguistics

The field of computational animal linguistics, while offering novel opportunities to explore the nuances of non-human communication, also faces an array of challenges, spanning from technical limitations in data acquisition to fundamental questions regarding the nature of language, to the boundaries of intelligence, and to the ethical responsibilities of researchers towards the subjects of their investigations.

6.1 Data Collection and Validation

Computational models are heavily dependent on data quality and quantity [241]; however, animal vocalizations are often located in noisy environments, overlap across individuals, and require expert-intensive annotation with often inconsistent inter-annotator agreement [38, 45, 70]. Datasets continue to be biased toward common, easily observable calls, motivating standardized protocols, automated annotation, and evaluation designed to expose failures on rare classes [5, 79, 240]. There is currently no method to reliably transcribe animal vocalizations into a set of standard symbols or known systems to anchor interpretation [163]. Researchers rely on playback experiments [136] and downstream tasks such as context or state classification [92, 276] to confirm computational representations, while avoiding anthropomorphism and confirmation bias that overinterpret signals as human-like language [228, 271, 285].

6.2 Ethical Considerations

Ethical considerations in computational animal linguistics require careful experimental design to minimize disturbance and ensure that observed behaviors reflect natural communication. A central concern is that human presence can induce stress and systematically alter behavior; for example, studies on primates show that proximity to researchers leads to measurable shifts in vocalization patterns and social interactions [227], potentially biasing data. These effects are compounded by invasive or disruptive practices such as deploying recording equipment, which can interfere with habitats, disrupt daily routines, and inadvertently increase predation risk or suppress essential activities like foraging and parental care [38]; in avian systems, poorly planned placement of devices near nests has been linked to reduced nesting success. Playback experiments, a common technique in animal communication research, introduce additional concerns by masking natural signals, disrupting mating or foraging, and inducing stress responses not only in target species but also across the broader acoustic community, as demonstrated in studies on marmosets [38]. Accordingly, such methods should be used sparingly and with strict controls on volume, duration, and repetition. Beyond immediate effects, repeated exposure to humans or experimental stimuli can lead to habituation, reshaping natural behaviors and compromising future data collection, while artificial or unfamiliar environments may further distort communication patterns. Taken together, these considerations underscore the need for minimally invasive methodologies, careful control of experimental interventions, and study designs that prioritize animal welfare while preserving the integrity of natural communication systems.

6.3 Opposition from Other Fields

Opposition to treating animal communication as language arises across linguistics, biology, and neuroscience, each emphasizing fundamental differences between human and non-human systems. Linguists caution against imposing human frameworks, noting that key properties of language—such as productivity, recursion, and hierarchical syntax—are

largely absent or only weakly evidenced in animals [118], and that apparent similarities in areas like speech perception are limited or inconsistent [141, 236]. Scholars such as Chomsky argue that human language is uniquely characterized by recursive and generative capacities requiring specialized cognition [273], while more recent studies suggest that structured animal signals may not map cleanly onto linguistic constructs like phonemes or syntax [34], and can often be explained by simpler ecological mechanisms [232]. From a biological perspective, anatomical constraints in the vocal tract limit many species’ ability to produce human-like speech [185], and even vocal mimics such as parrots rely on fundamentally different mechanisms [263], while differences in social structure may reduce evolutionary pressures for language-like systems. Neurologically, humans exhibit specialized cognitive and memory capacities supporting hierarchical processing beyond finite-state systems [83], with only limited and debated evidence of recursion in animals [94], alongside genetic differences such as human-specific adaptations of FOXP2 and related genes [72, 142, 233]. Collectively, these perspectives underscore the need for caution in applying human-centered linguistic models to animal communication, even as ongoing research continues to explore whether alternative forms of complexity may expand current definitions of language.

7 Current and Future Trends

7.1 Trend Corpus Construction

To quantify broad publication trends, we construct a separate corpus from the studies selected for the narrative review elsewhere in this survey. We use OpenAlex [207], an open scholarly index containing publication and bibliographic metadata across disciplines. Rather than define the corpus using a fixed keyword query, which can be sensitive to terminology and disciplinary indexing, we screen an OpenAlex works snapshot and apply progressively stricter relevance filtering. The snapshot covers publications from 1950–2026, with coverage through June 2026.

We retain English-language scholarly records with titles across articles and proceedings papers, preprints, books and book chapters, datasets, dissertations, reviews, reports, standards, and records assigned OpenAlex’s “other” work type for downstream relevance screening. We exclude retracted and paratext records, apply publication-status filtering where metadata are available, and retain the most recent version of duplicate records. These structural filters leave approximately 144 million works.

We then apply a SciBERT relevance classifier [16]. To construct its supervision, we assembled a gold-labeled seed set from diverse Google Scholar keyword searches for animal communication and vocalization studies spanning taxa and research methodologies; two independent annotators with domain expertise screened the retrieved papers for relevance. The dataset was balanced between relevant and irrelevant papers, with negative examples including topically adjacent search results such as animal-health studies unrelated to communication and papers whose animal-related terms occurred only in algorithm or model names. SciBERT was fine-tuned using title, abstract, venue, keyword, and topic metadata. At the 0.45 threshold used for the reported analysis, the classifier achieves an F1 score of 0.96 on a balanced held-out set of 4,049 papers and retains 257,036 candidate works. The remaining candidates are verified using

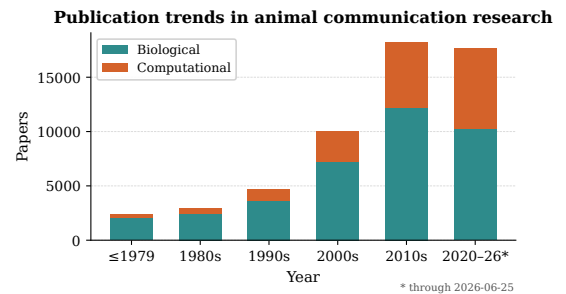


Fig. 5. Publication counts by research tradition. Pre-1980 years are grouped; the final bar (*) covers a partial period.

Table 4. Top 5 institutions and authors by research tradition (approximate publication counts).

Top institutions				Top authors			
Biological	Count	Computational	Count	Biological	Count	Computational	Count
Centre Natl. Rech. Sci.	1,000	Centre Natl. Rech. Sci.	511	Peter F. D. Boesman	256	James A. Simmons	212
Cornell University	929	Cornell University	495	Klaus Zuberbühler	167	Whitlow W. L. Au	206
University of St Andrews	708	University of St Andrews	423	Michael J. Ryan	156	James J. Finneran	174
Univ. Maryland, College Park	392	Scripps Inst. Oceanography	342	Robert J. Dooling	146	David K. Mellinger	137
Woods Hole Oceanographic Inst.	391	Woods Hole Oceanographic Inst.	332	Daniel J. Mennill	136	Rolf Müller	136

NVIDIA NemoTron-3-Nano-30B-A3B [25]. From each paper’s title and abstract, the model performs a stricter relevance check and extracts structured attributes including species, communication modality, features studied, application domain, research setting, and research tradition. This produces the final corpus of **55,854** papers used for the analyses below. We release the resulting corpus and the pipeline used to construct and analyze it in a public repository². Because OpenAlex is continuously updated, all statistics reported here refer to this frozen corpus.

7.2 Literature Trends

For the trend analyses below, 11 records with unrecognized research-tradition labels are excluded, leaving 55,843 papers. Of these, 37,712 are classified as biological and 18,131 as computational or hybrid. In the venue and species comparisons, arrows indicate rank relative to the corresponding ranking for the other tradition: ↑ denotes a higher rank, ↓ a lower rank, and – the same rank.

The clearest temporal trend is the recent acceleration of computational work (Figure 5), consistent with the increasing use of scalable machine-learning and audio-processing methods in bioacoustics. Despite this growth, computational animal communication has not formed a separate publication ecosystem and continues to share many publication venues with biological research, while its institutional and author distributions remain more distinct, suggesting a smaller and more specialized research community.

Several prominent venues appear in both traditions (Figure 7), including *The Journal of the Acoustical Society of America* (“J. Acoust. Soc. Am.”) and *Proceedings of the Royal Society B: Biological Sciences* (“Proc. R. Soc. B”). Taxonomic coverage likewise reveals a methodological bias: the overlap between the most studied taxa in the biological and computational traditions is concentrated in acoustically rich and readily recordable systems rather than representing a neutral sample of animal communication. The literature spans a range of application domains (Figure 6). Conservation and ecology is the largest

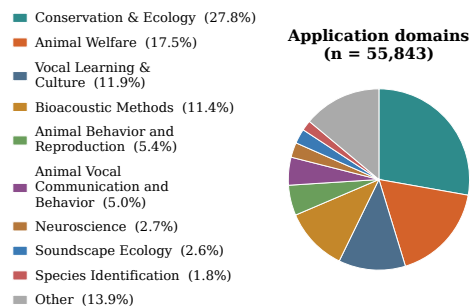


Fig. 6. Application domains in the curated corpus.

²<https://github.com/UTA-ACL2/cal-animal-comm-corpus>. The repository also provides the full NemoTron verification and metadata-extraction prompt and schema used to produce the structured corpus annotations.

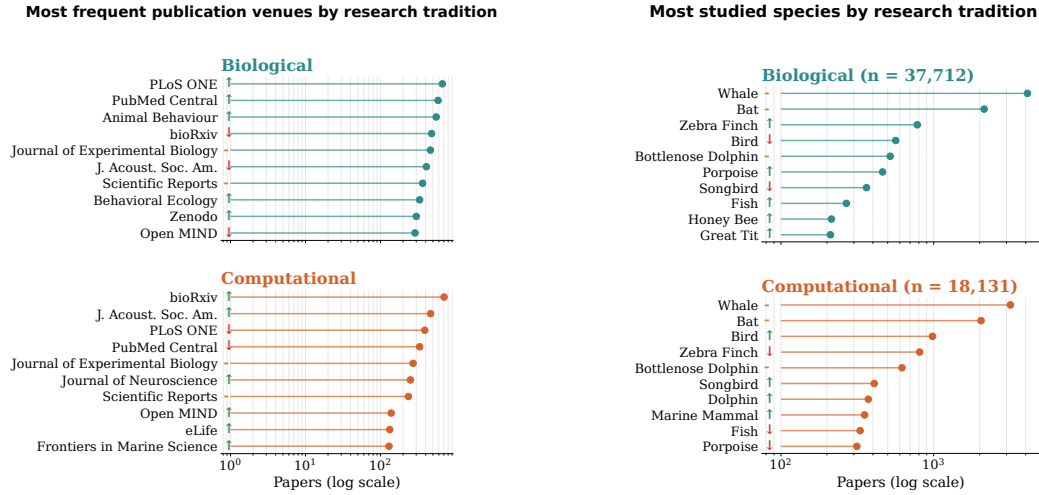


Fig. 7. Publication venues and studied taxa by research tradition. Left: top publication venues (log scale; abbreviated names). Right: most studied species by paper focus (log scale; top 10 per tradition), with whale, bat, and porpoise species grouped by genus.

category (27.8%), followed by animal welfare (17.5%), vocal learning and culture (11.9%), and bioacoustic methods (11.4%). These distributions show that animal communication research supports not only the study of communication itself, but also ecological monitoring, welfare, behavioral research, and methodological development. The two research traditions also differ in the communication features they study, although the Auditory/Acoustic Channel dominates both, accounting for 46.4% of biological and 46.5% of computational feature mentions.

The broader modality distribution shows an even stronger acoustic bias: 93.4% of extracted communication-modality mentions are Acoustic, compared with 2.9% Visual, 1.5% Chemical, and 1.0% Seismic. Chemical-communication papers are concentrated primarily in insects and other invertebrates, including honey bees and *Drosophila melanogaster*, while seismic-communication research is concentrated in African and Asian elephants, with additional work on insects and spiders. Recording settings are similarly unevenly represented (Table 5). Field and controlled laboratory studies together account for 84.9% of extracted setting assignments, while passive acoustic monitoring accounts for 8.4% and synthetic, captive, and citizen-science settings remain substantially less represented. Together, these trends indicate that computational animal communication is growing rapidly but remains centered on acoustic signals and a relatively narrow set of taxa and recording regimes. The distributions also highlight areas where current computational evidence is comparatively sparse, particularly non-acoustic modalities and less commonly studied collection settings.

Table 5. Research settings in the curated corpus ($n = 61,751$ assignments across 55,843 papers).

Setting	Assignments	Share
Field / Wild	29,006	47.0%
Lab / Controlled	23,404	37.9%
PAM	5,161	8.4%
Simulation / Synthetic	1,803	2.9%
Captive / Zoo	864	1.4%
Citizen Science	849	1.4%
Other	664	1.1%

7.3 Future Directions

Broader Data Coverage and Multimodality. The corpus reveals a strong concentration on acoustic communication and on field or controlled laboratory settings. This leaves comparatively less evidence for non-acoustic modalities, less-studied taxa, and collection regimes such as passive acoustic monitoring and citizen science. Future work should therefore prioritize broader taxonomic and ecological coverage, particularly for species and communication modalities that are poorly represented in current datasets. As larger datasets increasingly combine audio with video, spatial information, behavioral context, and other sensor streams, multimodal models may also provide a more complete account of communication than audio-only approaches [61, 261]. Importantly, multimodal systems should be evaluated not only on whether additional modalities improve predictive performance, but also on which contextual signals contribute useful information and whether those gains transfer across environments and species.

Generalization and Low-Resource Learning. A second priority is moving beyond within-dataset performance toward explicit evaluation of generalization. Animal communication datasets often differ substantially in recording devices, locations, background noise, individuals, and collection protocols, creating domain shifts that can substantially reduce model performance. The focal-to-PAM gap discussed in Sec. 3.1 is a clear example: systems trained on relatively clean focal recordings may not transfer reliably to passive soundscapes [109, 127, 211]. Future benchmarks should therefore report performance across individuals, sites, devices, recording regimes, and, where possible, species. This is especially important for data-scarce taxa, where collecting large labeled datasets may be impractical. Few-shot, zero-shot, self-supervised, and transfer-learning approaches are therefore likely to remain important, but should be evaluated under realistic distribution shifts rather than only on randomly partitioned examples from the same dataset [61].

Biological Validation and Evidence for Communicative Meaning. Improved predictive accuracy does not by itself establish that a model has identified communicative structure or meaning. Classification, clustering, tokenization, and generative modeling can reveal statistical regularities in vocalizations, but stronger semantic or linguistic interpretations require additional evidence. Future work should more closely integrate computational predictions with behavioral observations, controlled experiments, playback studies, and responses from animal receivers, following the evidence hierarchy in Sec. 3.4.1. Generative models raise a related challenge: acoustic similarity to natural vocalizations is insufficient if generated signals are not biologically plausible or do not elicit the expected behavioral response [272]. Progress in computational animal communication will therefore depend not only on larger or more capable models, but also on stronger out-of-domain evaluation, multimodal grounding, and experimental validation that connects computational structure to biological function.

8 Conclusion

The question of whether non-human animals possess language remains open, but this paper demonstrates that computational approaches are fundamentally reshaping how we investigate it. We formally introduce *Computational Animal Linguistics* and propose a five-stage pipeline—data collection, preprocessing, tokenization, meaning identification, and generation—that unifies the research process across species. Our survey of the thirteen linguistic properties reveals that animals exhibit far more meaningful content and structure than traditionally assumed and are worthy of further exploration. Our bibliometric analysis of domain-curated papers further confirms a marked terminology shift post-2010, reflecting the rapid absorption of deep learning and large language model techniques into the field. Case studies in songbirds, marine mammals, and canines illustrate how this pipeline already yields actionable results in welfare

monitoring, conservation, and human–animal interaction. Moving forward, the trends we have outlined demonstrate that this is not only an emerging field but an exponentially growing one, attracting greater interest over time. Future efforts will fill in current research gaps with higher quality data, generalizable models, and empirical validation. New discoveries will facilitate better communication and relationships between humans and animals; enable improved welfare for animals; advance large-scale data collection and crowd-sourced initiatives to support the research community; and foster collaborative efforts across multiple disciplines.

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Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009